

## CSC 380 HW#4 Setting up the Development Environment

1. Change your Google CoLab subscription to “Pro, free 1-year subscription for students”

### How to Sign Up for Free Google Colab Pro (Students & Faculty)

#### Eligibility

- Must be a **student or faculty member** at a **U.S. college or university**.
- Must be able to **verify academic status**, typically using a .edu email address.

#### Sign-Up Steps

1. **Visit the Colab for Higher Education Page**

Go to [Google Colab for Higher Education](#). [[blog.google](#)]

2. **Click the “No cost to students and educators” Button**

This will take you to the verification process.

3. **Verify Your Academic Status**

You’ll be asked to confirm your identity using your **institutional email address** (e.g., ending in .edu).

4. **Activate Your Free Colab Pro Subscription**

Once verified, you’ll receive access to Colab Pro for **one full year**, including:

- Faster GPUs
- Longer runtimes
- More memory
- Slideshow Mode for live code presentations

2. Create an OpenAI account and put a small amount of money (e.g. \$2) into the account.

### **Step 1: Create an OpenAI Account**

1. Go to:  <https://platform.openai.com/signup>
2. Sign up using:
  - Email + password OR
  - Google / Microsoft login
3. Verify:
  - Email address
  - Phone number (if prompted)

### **Step 2: Log in to the OpenAI Platform**

After signing up, go to:

 <https://platform.openai.com/>

This is your main dashboard.

### ● Step 3: Fix the “insufficient\_quota” Error (Critical)

If you see this error:

RateLimitError: insufficient\_quota (429)

👉 It means your account has **no available credits**

#### ◆ Step 4: Add a Payment Method

1. Go to: 👉 <https://platform.openai.com/account/billing>
2. Click: **“Add payment method”**
3. Enter:
  - Credit or debit card

#### ◆ Step 5: Add Funds / Enable Usage

After adding a card:

- Open: 👉 <https://platform.openai.com/account/usage>
- Then: 👉 <https://platform.openai.com/account/limits>
- Set:
  - **Soft limit:** \$3
  - **Hard limit:** \$5
  - ✓ This keeps your spending safe

#### 💡 Recommended: Add a Small Amount

- Add **\$2–\$5**
- This is enough for many assignments

#### ◆ Step 8: Wait and Retry

- Wait ~1–2 minutes after adding billing
- Run your code again
- ✓ The error should disappear

### 3. Obtain API access keys to OpenAI, HuggingFace and Langchain

#### 🔑 How to Obtain API Access Keys

##### 1. 🧠 OpenAI API Key

**Purpose:** Access GPT models and embeddings.

**Steps:**

1. Go to <https://platform.openai.com/signup> and create an account.
2. After logging in, visit <https://platform.openai.com/account/api-keys>.
3. Click **“Create new secret key”**, name it, and copy the key.
4. You may need to enter your **credit card** information, so you know.

##### 2. 😊 Hugging Face API Token

**Purpose:** Access models and datasets hosted on Hugging Face.

**Steps:**

1. Sign up at <https://huggingface.co/join>.

2. Go to <https://huggingface.co/settings/tokens>.
3. Click **“New token”**, choose **read** access, and copy the token.

### 3. ✓ How to Get a LangChain API Key

1. Go to <https://smith.langchain.com> and sign up.
2. After logging in, navigate to your **settings** and find the **API Keys** section.
3. Click **“Create API Key”**, name it, and copy the key.

## 4. Store API keys in CoLab “secret keys”

1. Open CoLab and click on the “secrets” button (a key-shaped) on the left.
2. Add a key/secret by specifying the name and your api key. One secret at a time.
3. Make sure the names are **“openai\_api”**, **“HF\_TOKEN”** and **“langchain\_api”** (to go with the start-up code).

The screenshot shows the Google Colab interface for a notebook named 'hw4\_1lm\_rag-starter.ipynb'. On the left sidebar, the 'Secrets' panel is open, displaying a table of stored secrets:

| Notebook access          | Name       | Value | Actions |
|--------------------------|------------|-------|---------|
| <input type="checkbox"/> | HF_TOKEN   | ..... |         |
| <input type="checkbox"/> | langchain  | ..... |         |
| <input type="checkbox"/> | openai_api | ..... |         |

Below the table, there is a '+ Add new secret' button and a 'Gemini API keys' dropdown. A text box explains: 'Access your secret keys in Python via:' followed by a code snippet:

```
from google.colab import userdata
userdata.get('secretName')
```

The main code area contains two Python code blocks. The first block imports necessary libraries for LangChain and HuggingFace:

```
[ ] import os, re, shutil
import requests
import chromadb
from typing import List, Dict, Tuple
from IPython.display import Markdown, display

# LangChain imports
from langchain.schema import Document
from langchain.document_loaders import PyPDFLoader, TextLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain.vectorstores import Chroma
from langchain_huggingface import HuggingFaceEmbeddings
from langchain.chains import LLMChain
from langchain_openai import ChatOpenAI, OpenAI
from langchain.prompts import PromptTemplate
```

The second code block shows how to load the secrets from the Colab interface and set them as environment variables:

```
[ ] # Load API keys from CoLab secret keys
from google.colab import userdata

langchain_api = userdata.get('langchain_api')
openai_api = userdata.get('openai_api')
hf_token = userdata.get('HF_TOKEN')

os.environ["OPENAI_API_KEY"] = openai_api
os.environ["LANGCHAIN_API_KEY"] = langchain_api
os.environ["HUGGINGFACEHUB_API_TOKEN"] = hf_token
```